

A photo you can't find is a memory you've lost

1.5B+

people use Google Photos each month

9T+

photos and videos stored

370M+

people search their photos each month (1 in 4)

150M

Google One subscribers paying for storage

More people finding the photo keeps the library worth paying for

Our goal: more users retrieve a photo they remember but can't precisely describe. **Why Google cares:** a library people can't search is worth less to keep, and storage is what Google One sells.



It becomes one number: User Retrieval Rate (URR)

Of the people who search for a photo they can only vaguely describe, the share who reach it within 28 days. It counts people, not searches.

$$\text{URR} = E \times [1 - (1 - I \times S \times R)^n]$$

E Express it

Can they put the memory into words at all?

Today: a search box, or a question to Ask Photos

I Be understood

Does Photos read the clue the way they meant it?

Today: matches people, places and things

S See it surface

Is the photo in the results at all?

Today: a grid of matching photos

R Recognise it

Can they tell it is the right one?

Today: thumbnails with date and place

n Try again

How many tries before they give up?

Today: retype the search

How to read it: a person succeeds only if they can express the memory and one try passes I, S and R together. One weak step caps the whole number, so the next slides find which step breaks.

We turned public complaints into a structured map of failed retrievals

1 · COLLECT

85,140

Public posts gathered (Play Store, App Store, YT, Reddit)

2 · FILTER

819

Relevant posts describing real photo search failures

3 · EXTRACT

144

Detailed retrieval attempts parsed into memories & queries

4 · AUDIT

203

Cases independently checked blind by a 2nd AI model

5 · PRIORITIZE

9 Areas

Problem areas ranked strictly where both models agreed

What We Analyzed	The Core Product Question Answered	Verification & Reliability
Photo Type	What kinds of photos do people struggle to retrieve?	Verified across store reviews
What Was Remembered	What details do people actually retain in memory?	High model agreement (cues)
What Was Forgotten	What details are gone when search fails?	High model agreement (lost info)
How They Searched	What queries do they type when memory is incomplete?	85.2% verified verbatim quotes
Where Search Broke	At what exact stage does retrieval fail?	Cross-model consensus (audit)

Why this is more than summarizing reviews

1. Objective data, not sentiment: We converted messy user rants into structured records with standardized fields, rather than subjective sentiment scores.

2. Hypotheses set before analysis: Potential failure causes were locked in advance so data couldn't be cherry-picked to confirm our biases.

3. Independent cross-model audit: A second AI model family re-audited the data blind. When models disagreed on theories, we dropped the theories and relied strictly on the failure points both models confirmed.

retrieval-discovery-engine.vercel.app

Sources: 85,140 total posts · 86.2% Play Store · 85.2% quotes verified in source text · Blind audit n=203 (κ 0.509 agreement on failure stages, Jaccard 0.673 on lost cues)

People remember the episode, but Search needs the date

Both data sources tell the same story: time is kept, the date is lost

What we looked at	Public Complaints (144 Real Searches)	User Interviews (15 App Users)
What was lost?	Personal life photos (80), trips (26), and videos (16)	Everyday photos; greatest urgency was lost medicine/documents (3/3)
What they remember	Rough time & season (40), objects (17), who was there (10)	Rough time & event (8), objects (7), people with them (6)
What they forget	The exact calendar date (37) and album names (29)	When it was taken (9) and exact search words (9)
What they actually do	20 of 32 typed just 1 bare word ("dog", "wedding")	9 of 15 gave up on search and manually scrolled the timeline

Rough time and events are the #1 solvable memory search ignores

42 users remembered a rough time or event: They had a clear life memory ("*last Diwali*", "*sister's graduation*"), but Search couldn't use it.

47 users had no memory clue at all: They had no searchable details retained.

The Insight: Users with a rough time represent the **single largest group of people who actually remember something** that current Search completely fails to use.

Where search breaks: 77% fail before any retry

Where the search broke	Share
1. Never surfaced: Exact date/keyword filter hid the photo	41.7% (60)
2. Clue misread: System misunderstood the rough memory	35.4% (51)
3. Couldn't refine / retry: Users gave up before retrying	2.8% (4)

✓ **Root Cause (Episodic Time):** People remember life moments ("*sister's wedding*"), but Search is built for calendar dates.

✓ **Secondary Gap (Visual Recognition):** When results appear, people can't verify them without seeing the surrounding moment (6 of 14 ended unsure).

✓ **What Was Overturned:** We initially thought users needed a "retry assistant"; the data proved search fails much earlier—by never showing the right candidates in the first place.

When they can't find it, people scroll—and nearly half still leave unsure

User survey: 15 complete retrieval stories

Methodology (n=15 users): structured questionnaire capturing exact memories, queries typed, and drop-off points.

Why: public posts lack query details; survey yielded 100% complete failure journeys.

Scope: qualitative directional sample showing real search habits.

[Survey questions ↗](#)

Survey confirms public complaint patterns		
Metric	Public Complaints (144)	User Survey (15)
Top memory cue	Time / season (28%)	Time / season (53%)
Ended unsure / empty	Not measurable	7 of 15 (47%)
Fell back to manual scroll	Frequently cited	9 of 15 (60%)

It costs most people over five minutes

Trips, LinkedIn photo, medical bill. **12 of 15 spent 5+ minutes** (exceeding brief's 5-min threshold); 3 had to ask someone or get the document again.

4 of 5 kept roughly when; only 1 put it in the search

ID	Target Photo	What They Remembered	What They Typed	Where Search Failed	Outcome
S04	Wedding photo	Roughly when, people	"wedding, pichle saal diwali"	Language / Slang	? Unsure
S03	Gym selfie	Roughly when, object	"gym"	Visual Clutter	? Unsure
S08	Trip photo	Roughly when, event	(Didn't know what to type; scrolled)	Vocabulary Block	✓ Found
S12	Family event	Roughly when, event	(Never typed; opened album)	Vocabulary Block	? Unsure
S06	Medicine bill	Object, text in bill	"medicine, bill"	Not Surfaced	✗ Never found

"wedding, pichle saal diwali" — S04 (Hinglish: a festival and relative year, not calendar date)

"medicine, bill" — S06 (Never found; had to get the document again)

Ask Photos feedback (4 of 4 stuck): 7 of 15 hadn't heard of it; **all 4 who tried it got "related photos, but not the one"**—leaving them stranded with no episodic trail.

Our target user remembers "when," but not the calendar date

Defined by what they remember, not by age

Long-time Google Photos users searching for a personal moment they can place only roughly.

They remember "around Diwali", "my sister's graduation", "last winter".

Segment Sizing Filter	Public Complaints (144)	User Survey (15)
Recalled rough time or event	42 (29%)	8 (53%)
Target Segment (After Exclusions)	37 (26%)	8 (53%)
Personal photo or video	30 of 37	All 8
Large library (5,000+ items)	Not measured	5 of 8

42 → 37: minus 2 who knew the exact calendar date (standard search works) and 3 general UI navigation complaints.

Why not an age band? Research proves memory state and library size drive search failure, not age. We use age as a diagnostic cut in testing, not as a product targeting rule.

Memory stores episodes; Photos stores items and dates

Human Memory Stores EPISODES

Rough time, event anchors, surrounding scenes ("last winter", "cousin's wedding")



Google Photos Indexes ITEMS & DATES

Isolated photos with calendar timestamps (YYYY-MM-DD) and tags

"I'll type in something super simple, like 'Halloween 2024' and it seriously can't find anything?" — Play Store review

5-Whys Root Cause Analysis (37 Segment Attempts):

1. Why isn't the photo found? It never surfaces in results (18 of 37).
2. Why doesn't it surface? Search misreads cues (12): "last Diwali" is neither a keyword nor a date.
3. Why do users search this way? Users remember the event, but lose the calendar date (#1 lost cue, 37 of 144).
4. Why can't Photos resolve it? Photos indexes isolated timestamps, not life episodes.
5. Root cause: Nothing converts rough memories into date windows and moment clusters.

Real app observation: 4 of 5 users typed bare nouns ("vacation", "gym"); 0 of 5 found the photo.

Segment: 37 public complaint attempts (26%), 8 of 15 in survey · Hinglish support is a design choice, not a finding

5 / 10

We recommend helping users recognise the moment—not write a better query

The problem, in the user's words

"I remember the event and roughly when ("last Diwali"), but Photos wants a date or keyword I've forgotten. So I scroll by hand, and even then I'm not sure the photo I found is the one."

We read this as a memory-to-recognition problem, not a generic search-quality problem.

The workaround is the design blueprint

Public complaints: 15 of 21 workarounds are manual scrolling (7 of 10 failed).

Survey: scrolling was first move for 9 of 15; 5 of those 9 still ended unsure.

Photos should take this step: convert rough time into a date window and group moments around it.

Our decision: bridge the memory-to-recognition gap, not redesign Search for every query.

Why scrolling fails: 5,000+ photo libraries cause recognition fatigue; users give up.

Each option helps, but none is built for "I'll know it when I see it"

Option	Good at	Role & gap for a vague memory
Photos search	People, places, things, dates	Needs exact terms or dates
Ask Photos (alone)	Conversational questions	All 7 who tried: "related, not the one"
Apple Photos	Describing the photo	Items, not moments or time
Memory Trails	Moments, inside Ask Photos	Recognise, then recover the moment

Ask Photos handles asking; **Memory Trails** is the recovery layer inside it.

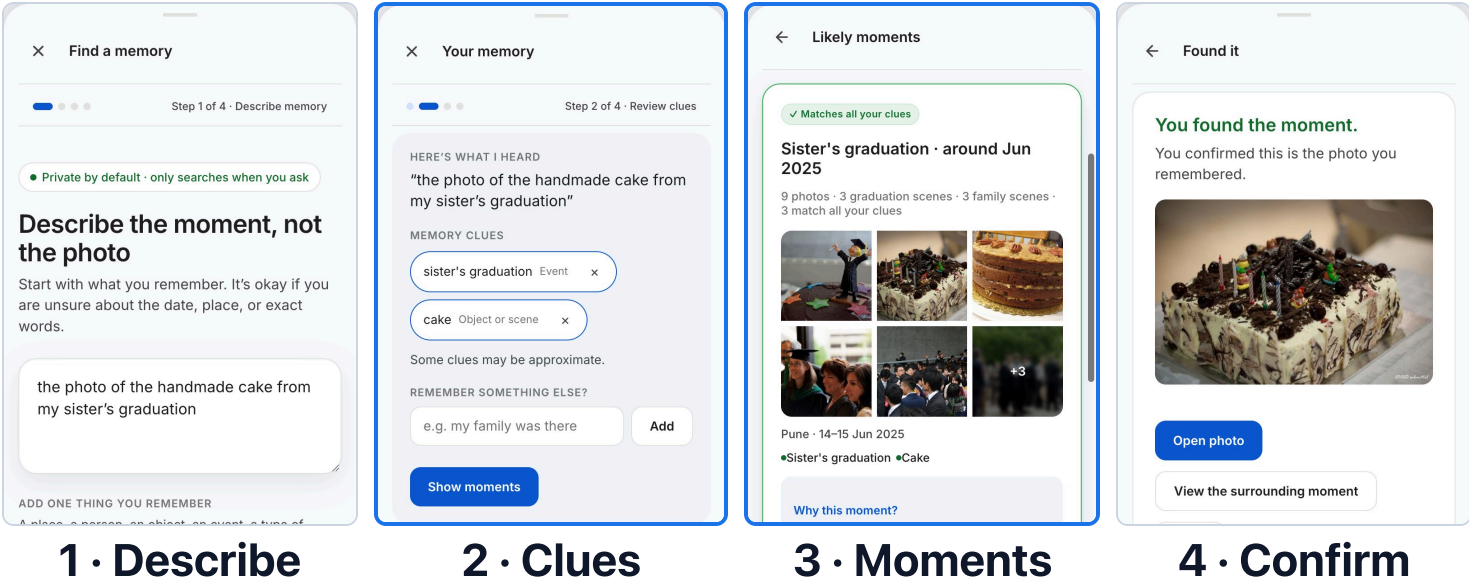
Hypothesis **Why Google wins:** personal-library signals plus recognition-first UI.

Five times the evidence changed the plan:

1. Metric: search success → **user URR** **2. Scope:** recovery agent → **moments browse** **3. Theory:** retry assistant → **surfacing audit** **4. Action:** retry search → **manual scroll** **5. Root:** "search broken" → **moment gap**

Memory Trails turns a vague memory into a trail of recognisable moments

Built inside Ask Photos
Integrated entry point: When Ask Photos results are vague, it prompts: *"Can't describe the photo? Try a Memory Trail"*, pre-populating clues.



memory-trails-v2.vercel.app

Blue frame = AI step

- Resolving search breakdowns:**
- **Vague time:** Converts relative phrases (*"pichle saal diwali"*) into date windows
 - **Mixed clues:** Extracts editable chips; user stays in complete control
 - **Visual clutter:** Clusters into episodic moments, not an endless flat grid
 - **Explainable AI:** Explains match reasons; user confirms final photo
 - **Wrong trail:** Offers nearby episodic moments with 1-tap undo if mistaken

Big gain with rich memory, modest with one vague cue Modelled

Memory detail	Query elements	Standard	Trails: Photo	Trails: Moment
Rich (3+ cues)	Time, place, activity	17.2%	96.2%	96.7%
Moderate (2 cues)	Scene + 2 vague clues	20.9%	27.6%	33.3%
Vague (1 cue)	Scene + rough timeframe	24.2%	30.9%	33.3%
No context (0 cues)	Scene description only	31.2%	31.2%	26.7%
Real-world weighted	Across 144 user attempts	25.9%	33.4% (+7.5%)	33.8% (+11.3%)

Core finding: Rich memories jump from 17% to 96%. For typical vague queries, **clustering into moments (+11.3% boost)** provides far greater practical value than re-ranking individual photos.

Scope: Evaluated on 1,282 labeled test photos. Focuses on the episodic trail interface inside Ask Photos, not a full Google Photos backend redesign.

Memory Trails MVP helped users find moments—but compound time still breaks it

ID	Own Photos search (baseline)	MVP Task 1 (cat)	MVP Task 2 (dog)	Sureness	vs GP	Adoption
R01	"Wedding before COVID" → wrong years	✓ Found, sure	✗ Not found	5/5	5/5	Every time
R02	"gym" → can't tell which	✓ Found, sure	✓ Found (1-tap alt)	4/5	4/5	When fails
R03	(Apple Photos user)	✓ Found, sure	✓ Found (nearby)	5/5	—	Every time
R04	"vacation" → wrong years	✓ Found, sure	✓ Found (1-tap alt)	4/5	4/5	When fails
R05	"Wedding event" → can't tell which	✓ Found, sure	✓ Found (clue edit)	4/5	4/5	Every time
R06	"Kerala photos" → can't tell which	? Right day, unsure	✓ Found (1-tap alt)	4/5	3/5	Every time

6 of 6

Fit the segment

4.33 / 5

MVP Task 1 sureness

4.0 / 5

MVP ease vs Photos

0 of 5

Found it in own Photos

MVP test questions ↗

Try Memory Trails MVP ↗

Testers valued Memory Trails' moments over a grid

- "Shows the moment, not a grid" — R02
- "Moving to nearby months, like scrolling but faster" — R03
- "Works without knowing what to type" — R04
- "Finds exact month & year... understands mix of Hindi and English" — R06

From these six tests, we will fix three things next

- Compound time:** R01 stuck on "pichle ke pichle saal" (year before last). *Fix:* parse compound offsets; editable year.
- Clues missed:** R04 didn't notice them. *Fix:* high-contrast "Showing Diwali 2025 · Tap to change".
- Speed:** R05, R06 wanted faster loading. *Fix:* pre-cache moments.

We will test whether Memory Trails increases confirmed retrieval

One number: users who reach the photo in 28 days **Proposed**

Definition: share of users with a vague-memory search in 28 days who reach the photo. Counts users, not searches.

Baseline 40% → target 45%. The baseline is a self-reported proxy (6 of 15 sure they found it). +5 pts is the smallest lift worth shipping.

We propose an 8-week user-level A/B, not yet run: ~1,600 users per arm for 80% power at α 0.05.

What the MVP has shown so far (not the A/B)

Modelled right photo in top 20: **25.9% → 33.4% (+7.5%)**; right moment in top 5: **33.8% (+11.3%)**

Proxy 6 testers' "sure" score **4.33 / 5**; 0 of 5 found target in own Photos search

We decide now when to ship, extend or stop **Proposed**

Ship if URR rises ≥5 pts and no guardrail trips. **Extend** if the lift is positive but under 5. **Stop** if false confirmation passes 3%.

Three numbers decide it **Proposed**

Role	Metric	Target
Primary	Confirmed retrieval (URR)	40% → 45%
Guardrail	False confirmation: said "found it", then kept searching	<3%
Experience	Time to first moment (95th percentile)	<1.5 s

Secondary signals, to explain any movement

Entry after a failed search (≥12%) · clue edits (15–25%) · top card matches all clues (rising) · confirm after opening a moment (≥55%) · "Not in any of these?" (falling) · abandonment (no rise)

The A/B has not run yet. Our evidence supports the direction but not a production-scale lift. We will test whether Memory Trails lifts *confirmed* retrieval without raising false confidence.

The opportunity is real, but we need one controlled test before scaling

Each risk has a mitigation built or planned

Risk	Mitigation & evidence
R1. The win sits where memory is richest (+79% rich, +7% vague)	Lean on moment recognition (moment in top 5: 33.3% for vague cues); split in A/B.
R2. Date filters hide photos	Soft scoring built: wrong date lowers rank, never hides (retrieval: 85% → 96%).
R3. False confirmation (worst failure)	Never auto-confirm; show surrounding photos; stop launch if false confirmation ≥ 3%.
R4. Can't read text in photos (0% recall on receipts)	Add OCR text indexing for screenshots, documents, and receipts.
R5. Ask Photos may already suffice	Ask Photos handles conversational search; all 7 testers got "related, not the one". Memory Trails completes it with moments.
R6. Sensitive memories (medical, relationships)	User-initiated only; session persists nothing; zero library data stored.

Pre-Launch Decision Matrix (Quality Gates):

● Ship (100%): URR lift ≥ +5% & false confirmation < 3%

● Iterate: lift +1% to +4.9%

● Rollback: false confirmation ≥ 3%

Known limitations & proxy boundaries

- Sample limits: 86.2% Play Store posts; convenience survey (n=15); 6 unmoderated tests.
- Model & telemetry: No internal telemetry; synthetic eval library (1,282 CC photos).
- Language signal: Hinglish support is a design choice, not a validated finding.

Observed user fixes from testing

1. Compound dates: parse offsets like "pichle ke pichle saal".
2. Clue clarity: make the active date chip prominent and editable.
3. Speed: pre-cache episodic moments; native gestures (pinch/swipe).

Our next step, and what we ask for

Pilot scope: 8-week user-level A/B test (~1,600 users/arm) in 1 market inside Ask Photos to validate confirmed retrieval lift.

We ask for approval to implement the three fixes and launch the controlled A/B test.